

Jugular Venous Doppler Ultrasound as a Non-Invasive Window to Right Heart Hemodynamics: A Narrative Review and Future Perspective

Ther Adv Pulm Crit Care Med

Volume 20: 1–14

DOI: 10.1177/
29768675251359700

© The Author(s) 2025.
Article reuse guidelines:
[sagepub.com/journals-](https://sagepub.com/journals-permissions)
[permissions](https://sagepub.com/journals-permissions)

Jon-Emile S Kenny , Philippe Rola, Ross Prager and Korbin Haycock

Abstract

The Venous Excess Ultrasound Score (VExUS) has produced great interest in **venous Doppler ultrasound** as a noninvasive means to **evaluate right heart hemodynamics**. While this score includes Doppler morphologies from sub-diaphragmatic veins, the **physiology of transcutaneous venous Doppler velocimetry and its change with cardiac pathology** was first studied and **described in the internal jugular vein (IJV)**. Over 50 years ago, the **systolic and diastolic velocity waves of the IJV were found to describe the x'- and y-descent of the jugular venous pulse (JVP)** in sickness and in health. Therefore, it is established that **abnormalities in right heart filling and function are reflected in the jugular venous flow velocity (JVfV) profile**. In this narrative review, we highlight the physiology of the JVP, its relationship to right heart performance, and, accordingly, its connection with JVfV. Grounded upon decades-old, pioneering investigations, we briefly highlight JVfV in patients with post-cardiopulmonary bypass physiology, atrial fibrillation, pericardial tamponade, and pulmonary hypertension. We then describe a novel, wireless, and wearable Doppler ultrasound that continuously displays JVfV and consider how this device informs diagnosis and therapy of acute circulatory dysfunction. We touch on gaps in knowledge and suggest future avenues of inquiry with special attention paid to synchronous acquisition and interpretation of venous and arterial Doppler measures. We emphasize the clinical relevance of this technology and physiological framework, including acute, inpatient shock resuscitation, volume removal (eg, “de-resuscitation”), and the **possibility for chronic, outpatient monitoring**.

Plain Language Summary Title

Internal Jugular Vein Doppler to Assess Right Heart Function

Plain Language Summary

Venous Doppler ultrasound is increasingly used to infer right heart hemodynamics. While the venous excess ultrasound (VExUS) score combines venous Doppler from multiple abdominal veins, historically, the internal jugular vein initiated our understanding of right heart function assessed via Doppler ultrasound of a peripheral vein. We review the basics of the venerable jugular venous pulse examination and how this relates to the central venous pressure and jugular venous Doppler. Finally, we describe a new technology that facilitates internal jugular vein Doppler measurements.

Keywords: Internal jugular vein, Doppler ultrasound, VExUS, venous congestion, review, wearable technology

Received: 4 January 2025; revised manuscript accepted: 30 June 2025.

Correspondence to:

Jon-Emile S Kenny, Health Sciences North Research Institute, Sudbury, ON, Canada; Flosionics Medical, Toronto, ON, Canada.
Email: jon-emile@heart-lung.org

Jon-Emile S Kenny
Health Sciences North Research Institute, Sudbury, ON, Canada.

Flosionics Medical, Toronto, ON, Canada.

Philippe Rola
Division of Intensive Care, Santa Cabrini Hospital, Montreal, QC, Canada.

Ross Prager
Western University, Division of Critical Care Medicine, London, ON, Canada.

Korbin Haycock
Department of Emergency Medicine, Riverside University Health System Medical Center, Moreno Valley, CA, USA.



Introduction

Venous Doppler ultrasound is experiencing a *renaissance* in critical care medicine, largely in response to the recently developed **Venous Excess Ultrasound Score (VExUS)**, first described by our group.¹ The VExUS score quantifies “venous congestion” and comprises 4 veins: (1) the inferior vena cava, as well as (2) the portal, (3) hepatic, and (4) intra-renal veins. These veins were chosen based upon clinical experience in cardiac anesthesiology² and data from the congestive heart failure population elaborating the pathophysiology of cardiorenal syndrome.^{3–5} More specifically, the hemodynamic measure most associated with worsening renal function is elevated central venous pressure (CVP, ie, pressure in the great veins adjacent to the right atrium), not cardiac output or mean arterial pressure.^{3,6} Therefore, venous Doppler abnormalities transmitted down into the abdominal compartment were the focus of VExUS, and increasing VExUS score associates with kidney dysfunction in multiple populations.^{7–10} However, VExUS can be time-consuming, requires advanced bedside ultrasound training, and obtaining adequate abdominal windows can be challenging. Furthermore, the visceral focus of VExUS overlooks the clinical history of venous Doppler ultrasound.¹¹ The earliest applications of venous Doppler were not directed at the abdominal organs, but rather focused on the great veins above the right atrium, ie, the superior vena cava (SVC) and internal jugular vein (IJV).^{11–13} These great veins were originally of interest because of the venerable jugular venous pulse (JVP) examination and its known abnormalities in cardiac disease.^{12,13} The JVP is a window to the right heart, which normally keeps venous pressure near zero millimeters of mercury (mm Hg); abnormalities in peripheral venous blood volume and distribution, as well as biventricular load and function, are expressed by changes in the height and contour of the JVP.^{13,14}

In this narrative review, we will describe the relationship between the JVP and right atrial pressure trace and present a brief history of how the IJV was the original transcutaneous target of venous Doppler ultrasound rather than the abdominal veins. We will touch upon pioneering research over the last 60 years, elaborating the relationship between perturbed right heart function and the jugular venous flow velocity (JVfV) measured non-invasively by jugular venous Doppler ultrasound. Finally, we highlight new wearable, ultrasound technology that

continuously displays JVfV to nurses and physicians and describe how, with this new tool, clinicians may guide resuscitation and de-resuscitation in a manner that is grounded on individualized hemodynamic phenotypes.

The JVP and Flow Velocity

JVP

The origin of the JVP is the P_{RA} waveform (Figure 1), which is the same as the central venous pressure (CVP) trace. Understanding how P_{RA} changes across the cardiac cycle in health explains the normal and—when there is pathological right heart hemodynamics—abnormal JVfV.^{12,13}

As diastole progresses, P_{RA} gradually rises until atrial contraction, which generates the a-wave. On clinical examination, this is generally difficult to see in the healthy patient, but can become prominent with early RV diastolic dysfunction (ie, stiffening of the RV) causing a clinically significant increase in P_{RA} during atrial contraction.^{14,21} Atrial relaxation then transiently diminishes the P_{RA} , causing the clinically imperceptible x-descent.¹³ With the onset of mechanical systole, the tricuspid valve (TV) closes—observed as the isovolumic contraction (IC) notch in the P_{RA} trace.¹⁵ Immediately following the IC is the c-wave, which, in the JVP, is caused by either carotid pulsation (as described by Mackenzie, see Figure 1B) and/or a transient P_{RA} elevation from the TV bulging into the right atrium.²² After the c-wave is the x' (ie, x-prime) descent, which is normally the greatest fall in P_{RA} during the cardiac cycle, due to increased right atrial capacitance when a competent TV is pulled down towards the apex of the RV by normal systolic function.^{15,23–25} This piston-like “suction” effect was first hypothesized by Aristotle²⁶ and later by Purkinje in the mid-19th century¹⁵; it is a clinical transduction of the TAPSE, or tricuspid annular plane systolic excursion.¹³ Another complementary mechanism causing P_{RA} to fall during systole (ie, the x'-descent) is something analogous to the Monro-Kellie hypothesis of the cranial vault, but instead applied to the rigid pericardium. During systole, cardiac volume shrinks more than pericardial volume when blood is ejected into the pulmonary artery and aortic arch. This effectively creates a vacuum within the pericardial sac that is transmitted to the right atrium.²⁷ The x'-descent may be observed in healthy patients on

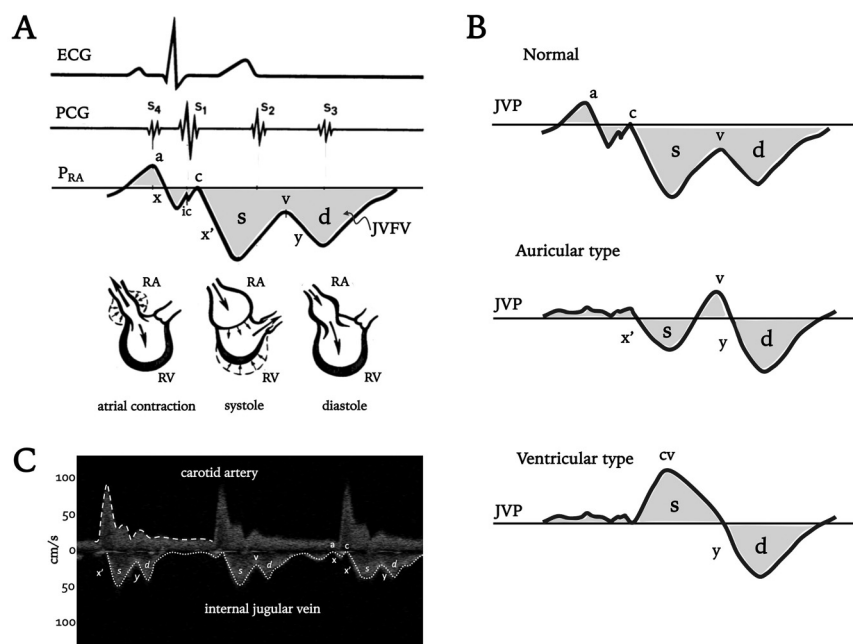


Figure 1. The physiology of the jugular venous pulse (JVP) and jugular venous flow velocity (JVfV). **(A)** Temporal relationship between the electrocardiogram (ECG), phonocardiogram (PCG), and right atrial pressure (P_{RA}). S_x of the PCG are the heart sounds. Increased P_{RA} or JVP pressure waves are the a, c, and v-waves described in the text. Decreased pressure waves are x, x', and y-descents. IC is the isovolumic contraction notch. The shaded grey region represents JVfV with s- (systolic) and d- (diastolic) waves. Cartoon showing right heart changes during cardiac cycle, RA is right atrium, and RV is right ventricle. See also.¹⁵ **(B)** Tracings of the JVP were first published in the mid-19th century¹² and more fully developed by Mackenzie at the turn of the century.^{16,17} The positive pulsations of the JVP were named after what were thought to be their underlying anatomical-physiological sources, with the a, c, and v-waves reflecting right atrium, common carotid, and right ventricle, respectively. Mackenzie also described the pathological progression of the JVP in tricuspid regurgitation from an initial a (auricular) type to v (ventricular) type.¹⁷ In early disease the prominent v-wave is due to atrial fibrillation (ie, “auricular”) because loss of atrial kick congests the right atrium and increases right atrial pressure (P_{RA}) during late systole as the competent tricuspid valve (TV) returns towards the base of the heart (ie, generating the tall v-wave). Later in disease, however, the v-wave becomes more pronounced and moves earlier into systole, reflecting regurgitant blood volume from the incompetent TV, so-called “ventricular” type v-wave prominence. A similar progression of the JVfV obtained by IJV Doppler ultrasound described by Ranganathan and Sivaciyan in patients with pulmonary hypertension is considered below.¹⁸ **(C)** Shows JVfV measured by wearable Doppler ultrasound also described below.^{19,20}

examination and is best seen between the 2 heads of the sternocleidomastoid muscle with tangential light; it is perceived as a very rapid loss of shadow timed with the radial pulse (ie, during mechanical systole), falling into the second heart sound (S_2)—the end of mechanical systole.^{12,13} P_{RA} rises again as the TV returns upwards towards the cardiac base with ongoing venous return; this pressure increase is the v-wave and is not seen in healthy subjects, but can be with disease, as described above (Figure 1B).¹⁷ When the TV opens, mechanical diastole begins and P_{RA} falls as venous blood deposits into the RV; this is the y-descent.²⁸ The y-descent, therefore, occurs

after the S_2 as the radial pulse recoils. It is not usually seen in healthy subjects but may be in hyperdynamic states; thus, a “double descent” seen as 2 rapid shadow disappearances on clinical examination should prompt further evaluation, such as Doppler ultrasound, described below.^{12,13} P_{RA} then rises again through diastole as the RA is charged with venous blood until the subsequent atrial contraction, or a-wave.

JVfV

Understanding the JVP is the foundation for interpreting the JVfV. Essentially, flow velocity in a great vein like the IJV is the “negative” of the

Table 1. Summary of central venous pressure, jugular venous pulse, jugular venous flow velocity, and the venous excess ultrasound scoring system.

	Advantages	Disadvantages
Central venous catheterization	When leveled correctly and measured at end-expiration and end-diastole, this is a direct measurement of right atrial end-diastolic pressure; waveform analysis clarifies right atrial hemodynamics.	Requires advanced training for placement and waveform interpretation; it is invasive with attendant risks of bleeding, infection, and pneumothorax; appropriate measurement conditions must be met.
Jugular venous pulse examination	Does not require specialized ultrasound equipment; can be used to estimate central venous pressure value (ie, as height of jugular meniscus above right atrium); vein surrounded only by neck soft tissue and atmospheric pressure.	Can be challenging to visualize in the neck; qualitative assessment that cannot be objectively displayed and viewed.
Jugular venous flow velocity examination	Requires only continuous wave Doppler ultrasound (ie, B-mode imaging not required); can be displayed and measured; can be used to estimate the height of jugular meniscus above right atrium; vein surrounded only by neck soft tissue and atmospheric pressure.	Requires specialized equipment and some training, especially if pulse wave Doppler ultrasound is used with imaging.
Venous excess ultrasound score	Non-invasive, multi-organ venous physiology assessment; good diagnostic specificity for predicting acute kidney injury in the critically ill and post-cardiac surgery populations.	Requires advanced training with pulse wave Doppler ultrasound, multiple abdominal vessels required to score; all veins affected by intra-organ and intra-abdominal pressures.

JVP. More concretely, when P_{RA} rises (eg, a, c, and v-waves), JVFV slows and is directed away from the right atrium (ie, towards the head), and when P_{RA} falls (eg, x, x', and y-descents), JVFV accelerates, towards the right atrium. Though there is nuance to the physiology of the JVFV considered below, in a healthy subject lying supine, it is normal to see a JVFV minimum near end-diastole (ie, corresponding to the a-wave), followed by a large increase in velocity during systole (ie, the s-wave). The s-wave is caused by both the x-descent (ie, atrial relaxation), but mostly the x'-descent (ie, the TAPSE) as the TV pulls the right atrium open like a piston. Another velocity minimum occurs with the v-wave as systole ends. Subsequently, the TV opens and the y-descent in pressure generates another velocity wave, the diastolic or d-wave. In this way, the JVFV is a Doppler ultrasound transduction of the P_{RA} trace (Figure 1C and Table 1).

Brief History of Venous Doppler Ultrasound

Clinical Doppler ultrasound was first used in the 1950s,²⁹ though its application to peripheral blood

flow analysis did not take shape for another decade with improved gain and filtering¹¹; the earliest peripheral vessels interrogated were the carotid arteries.¹¹ Then, with the development of bidirectional Doppler in the early 1960s, Kalmanson et al³⁰ first described venous Doppler in the IJV and related it to the P_{RA} and JVP. It is from their initial report on the jugular vein that *all* transcutaneous venous Doppler velocimetry is grounded. In addition to delineating the physiology of the JVP examination, the IJV was the target vessel of interest because of its relative ease-of-access; furthermore, pulsed-wave (PW) Doppler for directing ultrasound at deeper, internal vessels was not widely available.²⁹ In the ensuing years the group of Kalmanson and others elaborated upon the systolic and diastolic velocity waves (ie, “s” and “d” waves) seen in the IJV with various pathologies such as arrhythmia,²³ atrial and ventricular septal defects^{24,25} and they also described the contributions of the x and x'-descent to the venous Doppler s-wave.¹⁵ Following the initial descriptions of JVFV in the early 1970s,^{15,23–25,30–32} Sivaciyan and Ranganathan³³ offered the first classification

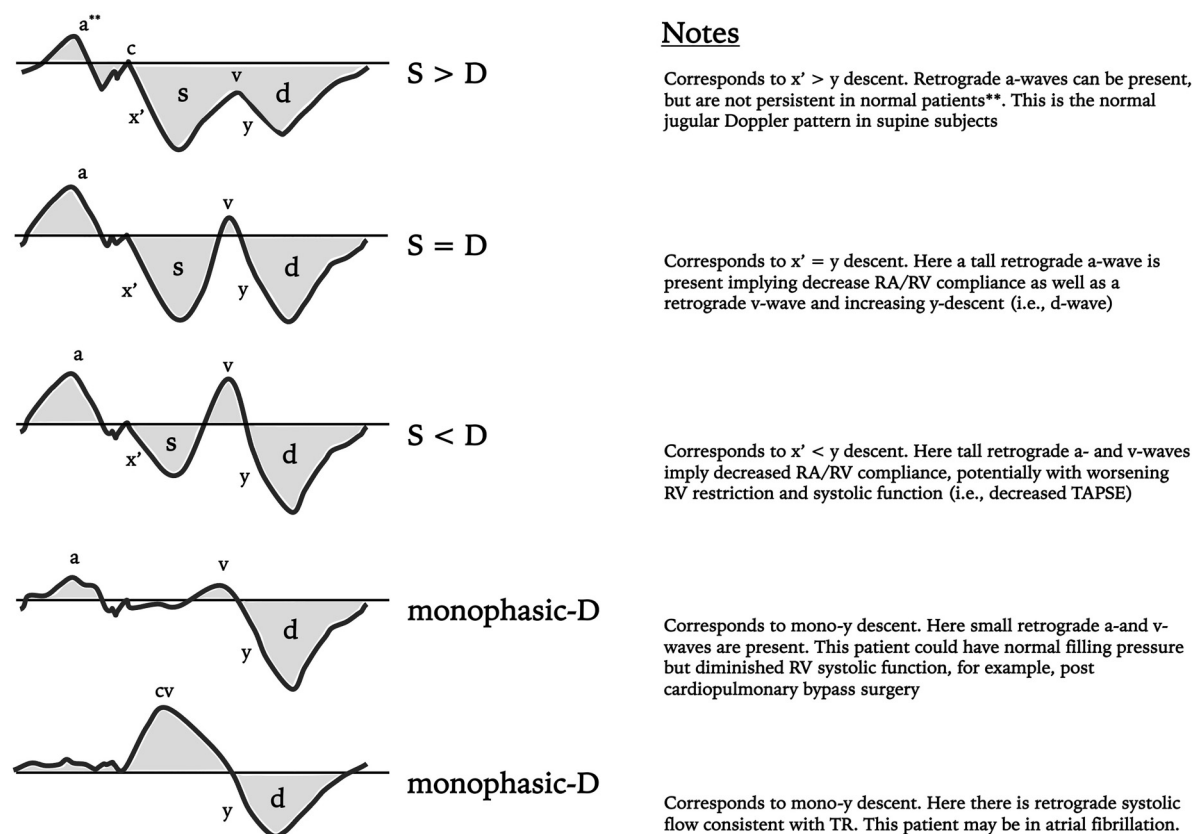


Figure 2. Diagram of jugular venous flow velocity (JVfV) classification system proposed by Sivaciyan and Ranganathan. See text for details. Here and in Figure 1, JVfV directed towards the right atrium is downwards on the y-axis, while velocity directed towards the head is upwards on the y-axis. In the supine position, the top pattern ($s > d$) is normal without persistent retrograde a-waves. Large retrograde a- and v-waves may be seen with the monophasic-diastolic pattern (not displayed here). Retrograde systolic velocity (bottom pattern) typically represents clinically significant tricuspid regurgitation (TR) but can occur in AV dissociation with atrial contraction against a closed tricuspid valve; in the latter, the retrograde systolic velocity is irregular. RA is right atrium; RV is right ventricle; TAPSE is tricuspid annular plane systolic excursion.

system for normal and abnormal venous Doppler morphologies based on the relative magnitude of the systolic and diastolic velocity waves. Their findings form the framework for normal and abnormal venous Doppler patterns in later studied veins (eg, hepatic,^{34,35} renal,⁴ common iliac,³⁶ and femoral^{37,38}).

Clinical Correlates

In 1978, Sivaciyan and Ranganathan³³ measured transcutaneous IJV Doppler in 174 cardiac patients (mostly ischemic and valvular disease). The JVfV was compared to invasively obtained right heart pressure in 82 patients. Additionally, the JVfV was related to the JVP and, in a subset, invasive SVC Doppler morphologies acquired

from the tip of a catheter introduced via the ante-cubital vein.

In the supine position, an s-wave greater than d-wave ($s > d$) was normal JVfV, with normal right heart measures and an x' -descent $>$ y-descent JVP (Figure 2). Invasive Doppler from the SVC showed the same pattern with only a slight transmission delay to the IJV. By contrast, $s = d$, $s < d$, and monophasic-d JVfV morphologies were all abnormal and corresponded to $x' = y$, $x' < y$, and solitary y-descent, respectively, in the JVP. All patients with these JVfV patterns had either elevated right heart pressures or were post-cardiac surgery patients. Of note, all patients with monophasic-d had elevated right heart pressures. The authors also considered the presence

of retrograde systolic JVFV to be abnormal, and these patients were found to have either A-V dissociation with cannon a-waves or, more likely, significant tricuspid regurgitation (TR). Lastly, end-diastolic retrograde flow in the JVFV (ie, from atrial contraction) was also an abnormal finding and corresponded to high a-wave pressure on right heart catheterization but the great majority also had $s > d$ JVFV suggesting that solitary, persistent end-diastolic retrograde flow could be an early sign of decreased RV compliance but with preserved RV function (ie, normal $x' > y$).

Brief elaboration of the JVFV in the following patient populations—post-cardiopulmonary bypass state, atrial fibrillation, cardiac tamponade, and pulmonary hypertension—is warranted, given additional research by Sivaciyan and Ranganathan, and others over the ensuing years.

Post-Cardiopulmonary Bypass

An explication of JVFV in post-cardiac surgery patients was carried out to determine why it was virtually universal for patients to transition from normal ($s > d$) to abnormal (ie, $s = d$ or $s < d$) following cardiopulmonary bypass despite ~40% having normal right heart pressure measured by catheterization.^{12,13,33,39} The authors concluded that decreased right atrial capacitance, perhaps from right atrial edema, scarring, or changes in pericardial constraint, decreased the x' -descent, increased the y -descent, or both. Of all variables studied, the only significantly different measure between patients with $s = d$ and $s < d$ morphologies following bypass was that the former had higher RV ejection fraction than the latter.³⁹ This follows the anticipated physiology given that $s < d$ signifies reduced RV function (eg, reduced TAPSE).¹³ Another possibility is altered properties of the pericardium following cardiac surgery. Part of the x' -descent is the fall in systolic pericardial pressure as the volume of the heart shrinks more than the pericardium when blood is ejected into the aortic arch.²⁷

Atrial Fibrillation

In patients with atrial fibrillation, the most common JVFV patterns were retrograde systolic flow (ie, due to TR), monophasic d-wave morphology, and $s < d$.^{18,33,39} The latter pattern illustrates that the loss of atrial kick does not completely abolish the RV Starling mechanism as some patients still had a perceptible x' -descent

(ie, s-wave). Without coordinated atrial contraction, the right atrium remains full at the expense of the RV. Consequently, with decreased RV preload, systolic TV excursion is diminished and $s < d$ is observed. These results were echoed in a follow-up study focused only on patients with pulmonary hypertension, where patients with atrial fibrillation displayed only $s < d$ or monophasic-d JVFV.¹⁸

Pericardial Tamponade

Pericardial tamponade characteristically displays very low jugular venous velocity that is confined to systole (ie, s-wave only) and during inspiration.³³ When there is tamponade physiology, the RV is restricted throughout diastole such that the y -descent is abolished. In general, it is very hard to measure JVFV during pericardial tamponade, though Sivaciyan and Ranganathan captured this data in 3 such patients. In accord with these observations, we have previously reported on JVFV during pericardial tamponade using a wearable Doppler device described below.⁴⁰

Pulmonary Hypertension

Finally, in pulmonary hypertension, ~90% of patients had some JVFV abnormality, including persistent, reverse end-diastolic flow (ie, retrograde a-wave), $s = d$, $s < d$, and monophasic-d.³³ The latter 2 abnormalities were observed in ~50% of the patients. A subsequent, observational study revealed similar JVFV abnormalities with clinical congestive heart failure only seen in patients with abnormal JVFV (especially $s < d$ or monophasic-d patterns).¹⁸ Progression of abnormal JVFV was also described in a patient over the course of a few years. Initially, there was $s > d$ with persistent retrograde a-waves, indicating decreasing right atrial and ventricular compliance (ie, early RV diastolic dysfunction) but with preserved RV systolic function (ie, normal x' -descent or s-wave). Over time, the y -descent increased and x' -descent decreased, leading to $s = d$ with worsening RV diastolic and systolic function. Finally, with RV dilation and TR, retrograde systolic flow with monophasic-d waves was seen.^{18,32} These observations are quite similar to the progression described by Mackenzie in 1907, but by using only the JVP (Figure 1B). In 2001, Matsuyama et al⁴¹ expanded upon the aforementioned by recording JVFV in 64 patients with chronic obstructive

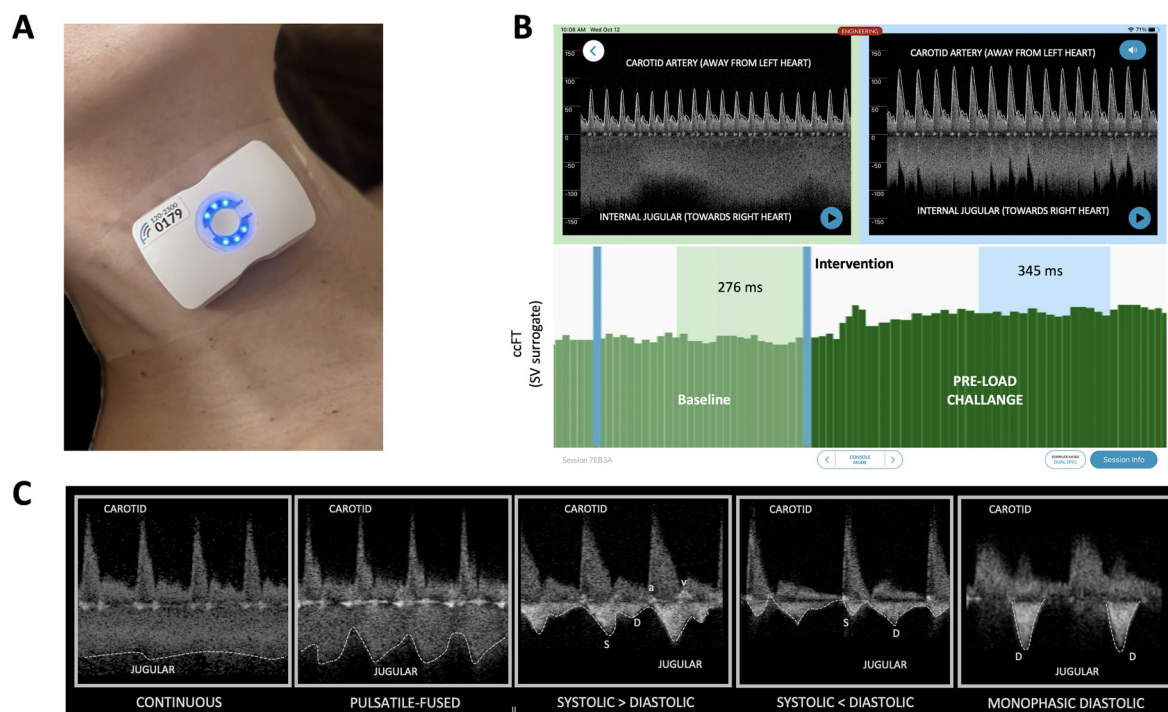


Figure 3. The wireless, wearable Doppler ultrasound: **(A)** the device on a healthy subject **(B)** the graphical user interface of the wearable Doppler displayed on any iOS system **(C)** jugular venous flow velocity patterns captured by the wearable Doppler ultrasound based upon the classification system of Sivaciyan and Ranganathan. Continuous and pulsatile-fused morphologies occur when the jugular vein collapse point is below or near the level of the wearable Doppler ultrasound, respectively. Systolic > diastolic (ie, $s > d$) morphology indicates that the jugular vein collapse point is above the wearable Doppler and the right atrial pressure waveform is transmitted to the level of the neck at which the ultrasound is adhered. The mechanisms of systolic < diastolic and monophasic diastolic are described in the text as well as in Figure 2 and Table 2. These Doppler patterns form the x-axis of the Doppler Starling curve shown in Figure 5.

pulmonary disease being worked up for pulmonary hypertension.⁴² The authors were especially interested in jugular venous Doppler because non-invasive echocardiographic methods to estimate pulmonary pressure can be limited by chest hyperinflation. They found that falling s-wave relative to d-wave (ie, x'-descent relative to y-descent) was directly proportional to mean pulmonary artery pressure (MPAP) measured by right heart catheterization in a strong, linear manner. Using $s < d$ as a cut-off (ie, a d-wave to s-wave ratio of at least 1.0), they found a 71% sensitivity and 95% specificity for diagnosing pulmonary hypertension (ie, MPAP of at least 25 mm Hg on catheterization). This ratio was more specific than echocardiography using TR jet to estimate pulmonary artery systolic pressure; furthermore, 88% of patients had an adequate echocardiography window, while all patients had an adequate jugular venous Doppler window.

New Technology and Future Investigation

A wireless, wearable, continuous wave Doppler ultrasound that insonates both the common carotid artery and IJV has recently been developed for resuscitation (Figure 3).^{19,20,43,44} One unique aspect of this device is that it continuously displays the JVFV as a surrogate for right heart hemodynamics in addition to carotid artery measures to infer left heart function.⁴⁵ We were the first to describe the synchronous assessment of venous and arterial Doppler measures as the "Doppler Starling curve," which is especially useful when performing a preload challenge such as a rapid fluid infusion, Trendelenburg maneuver, or passive leg raise (PLR).⁴⁶ For instance, we described a patient in whom serial fluid boluses improved forward arterial flow (as observed in the common carotid artery) but caused the JVFV to progress from $s > d$ to $s < d$, indicating early RV dysfunction with preload.⁴⁷ Conversely, we

Table 2. Differential diagnosis of altered jugular venous flow velocity. See also.^{12,13}**Diminished S wave relative to D wave**

1. Reduced right ventricular contraction
 - i. Right ventricular failure in pulmonary hypertension
 - ii. Post-cardiac surgery right ventricular dysfunction
 - iii. *Right ventricular infarction or acute pulmonary embolism*
2. Atrial fibrillation and loss of the Starling mechanism

Exaggerated D wave relative to S wave

1. Increased v-wave pressure without restriction to ventricular filling
 - i. High sympathetic tone
 - ii. *Hypervolemia*
 - iii. *Extra source of venous filling (eg, atrial septal defect)*
 - iv. *Pericardial effusion with some restriction (presence of y-descent or d-wave excludes tamponade physiology)*
 - v. *Constrictive pericarditis*
 - vi. *Pulmonary hypertension with elevated right ventricular diastolic pressure*
 - vii. *Right ventricular ischemia and/or infarction*
 - viii. *Cardiomyopathy*
2. Decreased right atrial capacitance following cardiac surgery

also showed that patients with normal baseline right heart hemodynamics per JVFV can rapidly develop venous hypertension with additional preload,^{47,48} a state termed “dynamic fluid intolerance,”^{49–52} given that physiological harm by preload is only evidenced by a dynamic maneuver like the PLR. As an expansion of the Doppler Starling curve framework, we have described Doppler ‘quadrants’ to phenotype hypo-perfused patients⁴⁸ with venous-arterial Doppler-enhanced resuscitation (VADER⁵³) to direct therapy. Recently, a team of highly trained nurses proficient in point-of-care ultrasound have integrated the wearable Doppler into their workflow for early sepsis care.⁵⁴ Nevertheless, the wearable Doppler ultrasound does require training as users learn how to obtain a good Doppler window for both the common carotid artery and internal jugular vein, ensuring adequate flow velocity spectrograms for both vessels. As well, the device, like any ultrasound system, is susceptible to patient motion and artifact (eg, vocalization).

The Importance of Gravitational Position and Starling Resistors

One important finding from the wearable Doppler is the presence of “continuous” and “pulsatile-fused” JVFV morphologies,⁵⁵ which were not described by Sivaciyan and Ranganathan in their original classification scheme, though these morphologies are observed in other veins such as the intra-renal vein.⁵ In healthy volunteers, we found the $s > d$ morphology to be normal, but only in the supine position. When healthy volunteers are upright or in semi-Fowler, a “continuous” or “pulsatile-fused” morphology is ubiquitously seen.⁵⁵ In one subject, we showed that the transition from “continuous” in the upright position to $s > d$ in the supine position occurred because of changes in the position of the jugular vein collapse point, which is a Starling resistor, or when an elastic tube varies its cross-sectional area as a function of its internal pressure relative to external pressure (ie, the transmural, or “across the wall,” pressure). When there is a jugular venous Starling resistor (ie, the point at which atmospheric pressure around the neck supersedes the jugular

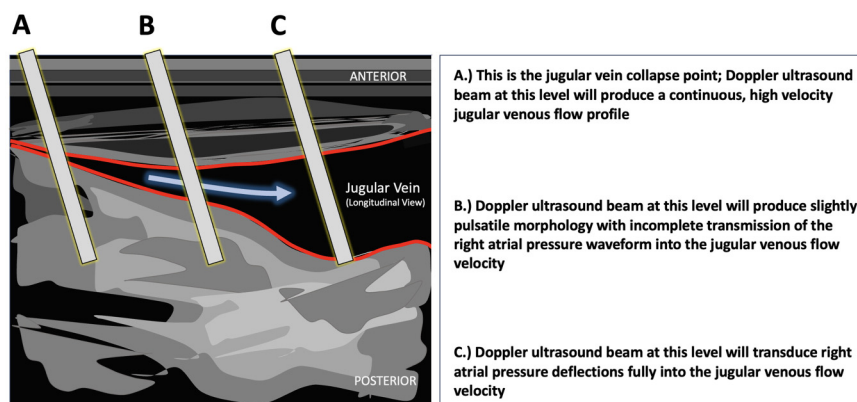


Figure 4. Illustration of the internal jugular vein in the longitudinal view. The internal jugular vein is outlined in red, and the direction of venous return from the head to the right heart is shown by the light blue arrow. Three different Doppler ultrasound beam levels describe the relationship to jugular venous flow velocity.

pressure), the jugular vein collapses, which is sometimes termed the jugular “meniscus” on b-mode imaging.^{56,57} This anatomy and physiology are commonly taught to appear like a “wine bottle” in the jugular vein; above the collapse point is the “neck,” while below the collapse point is the “body” of the wine bottle. When the wearable Doppler ultrasound insonates above the collapse point (ie, the bottle-neck), a “continuous” or “pulsatile fused” JVFV is observed, whereas if insonation is below the collapse point (ie, the body), then one sees the morphologies described by Sivaciyan and Ranganathan. In this manner, when a continuous morphology is present, the CVP can be assumed to be *less than* the height of the wearable Doppler above the right atrium. Conversely, if a pulsatile morphology is observed (eg, $s > d$, $s = d$, $s < d$, or monophasic-d), then the CVP can be assumed to be *greater than* the height the wearable Doppler is above the right atrium (Figure 4). Given this, when we dichotomized preload based upon JVFV into “low” versus “high,” we placed $s > d$ into the “high” category but only when the reading is made with a patient semi-recumbent or upright position.^{47,48} When supine, $s > d$ is normal. Furthermore, only the IJV reliably discerned high from low preload (based upon gravitational position) as compared to the hepatic, portal, and intra-renal vein morphologies in healthy volunteers.⁵⁵ Concordant with this finding, in a small case series of patients with left ventricular assist device (LVAD) who had IJV Doppler measured by the wearable ultrasound,⁵⁸ the one patient with a pulsatile IJV pattern required up-titration of diuretics within 6 weeks of assessment, while those with continuous morphology did not.

Future Directions

In terms of future research, the JVFV pattern lacks patient outcome data as VExUS does (eg, acute kidney injury risk, ICU length of stay, etc). Accordingly, we wonder if the JVFV pattern can predict organ dysfunction (eg, acute kidney injury, delirium, etc) better than the central venous pressure in post-cardiac surgery or septic patients? Might this apply to outpatients with disorders of blood volume (eg, chronic kidney disease, pulmonary hypertension, congestive heart failure, cirrhosis, etc)? It may be tempting to relate IJV morphology to P_{RA} as a research project.⁸ On face value, this may seem appropriate; however, caveats should be considered. For instance, changes in pericardial or right atrial capacitance (as occur in the post-cardiopulmonary bypass state) mean that an abnormal JVFV morphology can occur with normal right heart filling pressure, especially if there is reduced RV function, as Sivaciyan and Ranganathan have already established.^{12,13,33,39} This simply emphasizes the point that P_{RA} occurs at the crossroads of cardiac function and venous return.^{59,60} A euvoletic or hypovolemic patient with abnormal RV function could have a normal P_{RA} but an abnormal JVFV pattern. A better comparison with P_{RA} would be to establish the height above the right atrium at which a continuous JVFV transitions into discernible s- and d- waves. Another caveat, and in a similar vein, would be comparing JVFV to P_{RA} in the setting of high pericardial pressure that occurs when there is tamponade or high levels of positive end-expiratory pressure (PEEP). In this situation, there could be a very high P_{RA}

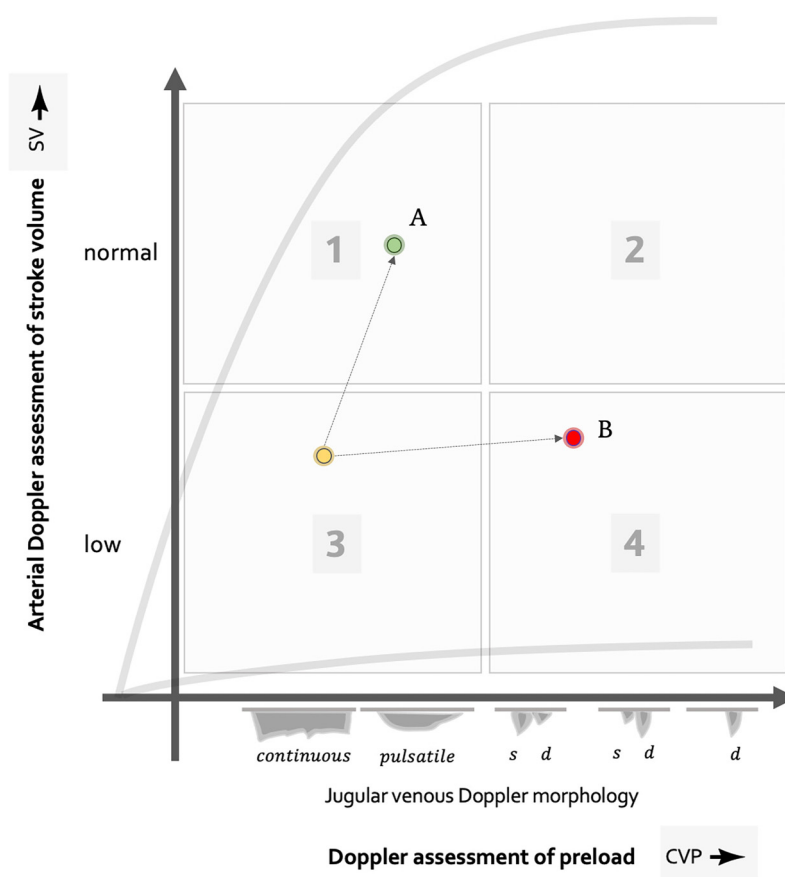


Figure 5. Illustration of venous-arterial enhanced Doppler resuscitation (VADER) and the Doppler Starling curve. In the baseline, semi-recumbent position, 2 patients can begin with a similar hemodynamic profile, indicating low/normal right heart filling assessed by jugular venous flow velocity (x-axis) and stroke volume (SV, y-axis); this is the yellow point in quadrant 3. The provision of intravenous fluids to these 2 patients leads to very different hemodynamic responses. Patient A has the desired response moving upwards to quadrant 1; that is, there is little change in right heart filling but a large increase in SV (green dot). By contrast, patient B has an undesirable response revealing abnormal right heart filling with little increase in SV, moving out to quadrant 4 (red dot). Patient B illustrates “dynamic fluid intolerance.”

relative to atmospheric pressure (to which transducers are leveled) but diminished right atrial *transmural pressure*, which is the P_{RA} relative to the pericardial pressure.⁵⁹ When the right atrium is restricted from without, the y-descent is abolished. In this regard, a high P_{RA} will not be seen in association with a rising d-wave relative to s-wave. Jugular venous velocities will be reduced because of jugular vein distension and only systolic velocities are present, generally when intrathoracic pressure falls (eg, during spontaneous inspiration).^{12,13,33} Finally, we propose that venous Doppler ultrasound to assess hemodynamics should always occur in the context of concurrently measured arterial Doppler ultrasound so as to integrate “congestion” and

“perfusion” metrics simultaneously.^{43,44,46,48,51,53} In this regard, future study of Doppler phenotypes akin to the original Diamond-Forrester framework⁵¹ will better direct diagnosis and therapy of acute and chronic circulatory failure (Figure 5). Multi-center, prospective studies relating Doppler phenotypes (ie, “quadrants”) to patient outcome are needed, and retrospective machine learning driven algorithm development to predict outcome from changes in the venous-arterial Doppler signals during sepsis resuscitation is underway.

Conclusion

Venous Doppler ultrasound is an exciting and emerging measure in acutely ill patients to assess

right heart hemodynamics. While veins below the right atrium have been of interest because of the VExUS score, the vein in which venous Doppler morphology was first studied and understood was the internal jugular vein; this is because of historical clinical interest in the jugular venous pulse. The systolic and diastolic velocity waves seen in the IJV follow the x'- and y-descents of the JVP (ie, the right atrial pressure waveform). Because it is established that JVFV mirrors right heart hemodynamics from pioneering investigations conducted over 40 years ago, JVFV can presently be integrated into clinical decision-making with simple, continuous wave Doppler ultrasound. New wearable technologies make JVFV more accessible to physicians and nurses who can incorporate this data into both the resuscitation and de-resuscitation of acutely and chronically ill patients. Physiological frameworks such as the Doppler Starling curve and VADER should be used to direct future investigation of JVFV patterns on patient outcome and inform interventional studies.

List of Abbreviations

VExUS	venous excess ultrasound score
IJV	internal jugular vein
JVP	jugular venous pressure
JVFV	jugular venous flow velocity
CVP	central venous pressure
SVC	superior vena cava
P_{RA}	right atrial pressure
TV	tricuspid valve
IC	isovolumic contraction notch
RV	right ventricle
TAPSE	tricuspid annular plane systolic excursion
S_2	second heart sound
RA	right atrium
PW	pulsed wave
TR	tricuspid regurgitation
MPAP	mean pulmonary artery pressure
PLR	passive leg raise
VADER	venous arterial enhanced Doppler resuscitation
LVAD	left ventricular assist device
PEEP	positive end-expiratory pressure

Acknowledgments

Not applicable.

Author contribution(s)

Jon-Emile S Kenny: Conceptualization; Data curation; Writing – original draft.

Philippe Rola: Data curation; Methodology; Resources; Writing – review & editing.

Ross Prager: Investigation; Methodology; Validation; Writing – review & editing.

Korbin Haycock: Formal analysis; Investigation; Resources; Writing – review & editing.

Availability of data and materials

Not applicable.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of conflicting interests

The author(s) declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: JESK is the co-founder and chief medical officer of Flosonics Medical; no other interests are declared.

Ethics approval

Not applicable.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Jon-Emile S Kenny  <https://orcid.org/0000-0002-3654-1146>

References

1. Beaubien-Souligny W, Rola P, Haycock K, et al. Quantifying systemic congestion with point-of-care ultrasound: Development of the venous excess ultrasound grading system. *Ultrasound J* 2020; 12: 1–12. doi:10.1186/s13089-020-00163-w
2. Tremblay J-A, Beaubien-Souligny W, Elmi-Sarabi M, et al. Point-of-care ultrasonography to assess portal vein pulsatility and the effect of inhaled milrinone and epoprostenol in severe right ventricular failure: A report of 2 cases. *AA Case Reports* 2017; 9: 219–223.
3. Mullens W, Abrahams Z, Francis GS, et al. Importance of venous congestion for worsening of renal function in advanced decompensated heart failure. *J Am Coll Cardiol*. 2009; 53: 589–596. doi:10.1016/j.jacc.2008.05.068

4. Iida N, Seo Y, Sai S, et al. Clinical implications of intrarenal hemodynamic evaluation by Doppler ultrasonography in heart failure. *JACC Heart Fail.* 2016; 4: 674–682.
5. Tang WW, Kitai T. Intrarenal venous flow: A window into the congestive kidney failure phenotype of heart failure? *JACC Heart Fail.* 2016; 4: 683–686.
6. Verbrugge FH, Dupont M, Steels P, et al. Abdominal contributions to cardiorenal dysfunction in congestive heart failure. *J Am Coll Cardiol.* 2013; 62: 485–495. doi:10.1016/j.jacc.2013.04.070
7. Argaz ER. VExUS Nexus: Bedside assessment of venous congestion. *Adv Chronic Kidney Dis* 2021; 28: 252–261. DOI: 10.1053/j.ackd.2021.03.004
8. Longino A, Martin K, Leyba K, et al. Correlation between the VExUS score and right atrial pressure: A pilot prospective observational study. *Crit Care* 2023; 27: 205. DOI: 10.1186/s13054-023-04471-0
9. Spiegel R, Teeter W, Sullivan S, et al. The use of venous Doppler to predict adverse kidney events in a general ICU cohort. *Crit Care.* 2020; 24: 1–9. doi:10.1186/s13054-020-03330-6
10. Turk M, Robertson T, Koratala A. Point-of-care ultrasound in diagnosis and management of congestive nephropathy. *World J Crit Care Med* 2023; 12: 53–62. DOI: 10.5492/wjccm.v12.i2.53
11. Veyrat C. Those who faced turbulence and launched the era of flow dynamic concepts for cardiac investigation. *Cardiovasc Res* 2001; 51: 205–216. DOI: 10.1016/s0008-6363(01)00307-8
12. Ranganathan N, Sivaciyan V. Jugular venous flow velocity pattern application to bedside recognition of jugular venous pulse contour and right heart hemodynamics. *Am J Noninv Cardiol* 1993; 7: 75–88. DOI: 10.1159/000470256
13. Ranganathan N, Sivaciyan V. Jugular venous pulse descents patterns – Recognition and clinical relevance. *CJC Open* 2022. DOI: https://doi.org/10.1016/j.cjco.2022.11.016
14. Magder S. Right atrial pressure in the critically ill: How to measure, what is the value, what are the limitations? *Chest* 2017; 151: 908–916. doi:10.1016/j.chest.2016.10.026
15. Kalmanson D, Veyrat C, Chiche P. Atrial versus ventricular contribution in determining systolic venous return. A new approach to an old riddle. *Cardiovasc Res* 1971; 5: 293–302. DOI: 10.1093/cvr/5.3.293
16. Mackenzie J. Pulsations in the veins, with the description of a method for graphically recording them. *J Pathol Bacteriol.* 1892; 1: 53–89. doi:10.1002/path.1700010107
17. Mackenzie J. The interpretation of the pulsations in the jugular veins. *Am J Med Sci.* 1907; 134: 12. doi:10.1097/00000441-190707000-00002
18. Ranganathan N, Sivaciyan V. Abnormalities in jugular venous flow velocity in pulmonary hypertension. *Am J Cardiol.* 1989; 63: 719–724. DOI: https://doi.org/10.1016/0002-9149(89)90258-0
19. Kenny J-ÉS, Munding CE, Eibl JK, et al. A novel, hands-free ultrasound patch for continuous monitoring of quantitative Doppler in the carotid artery. *Sci Rep.* 2021; 11: 1–11. doi:10.1038/s41598-021-87116-y
20. Kenny J-ÉS. Functional hemodynamic monitoring with a wireless ultrasound patch. *J Cardiothorac Vasc Anesth.* 2021; 35: 1509–1515. DOI: https://doi.org/10.1053/j.jvca.2021.01.040
21. Magder S. Central venous pressure monitoring. *Curr Opin Crit Care.* 2006; 12: 219–227. doi:10.1097/01.ccx.0000224866.01453.43
22. Rich LL, Tavel ME. The origin of the jugular C wave. *N Engl J Med* 1971; 284: 1309–1311. DOI: 10.1056/nejm197106102842306
23. Kalmanson D, Veyrat C, Chicke P. Venous return disturbances induced by arrhythmias. A transcutaneous, instantaneous, flowmetric study at the site of the jugular vein. *Cardiovasc Res* 1970; 4: 279–290. DOI: 10.1093/cvr/4.3.279
24. Kalmanson D, Veyrat C, Deraï C, et al. Non-invasive technique for diagnosing atrial septal defect and assessing shunt volume using directional Doppler ultrasound. Correlations with phasic flow velocity patterns of the shunt. *Br Heart J* 1972; 34: 981–991. DOI: 10.1136/hrt.34.10.981
25. Kalmanson D, Aigueperse J, Veyrat C, et al. Non-invasive technique for diagnosing congenital and acquired ventricular septal defects using directional Doppler ultrasound. Correlations with phasic flow velocity patterns of the shunt. *Br Heart J* 1974; 36: 428–445. DOI: 10.1136/hrt.36.5.428
26. Brecher GA. Cardiac variations in venous return studied with a new bristle flowmeter. *Am J Physiol-Legacy Content* 1954; 176: 423–430. DOI: 10.1152/ajplegacy.1954.176.3.423

27. Holt JP, Rhode EA, Kines H. Pericardial and ventricular pressure. *Circ Res* 1960; 8: 1171–1181. DOI: 10.1161/01.res.8.6.1171
28. Magder S, Erice F, Lagonidis D. Determinants of the Y descent and its usefulness as a predictor of ventricular filling. *J Intensive Care Med*. 2000; 15: 262–269. doi:10.1177/088506660001500505
29. Fraser AG, Monaghan MJ, van der Steen AFW, et al. A concise history of echocardiography: Timeline, pioneers, and landmark publications. *Eur Heart J Cardiovasc Imaging* 2022; 23: 1130–1143. DOI: 10.1093/ehjci/jeac111
30. Kalmanson D, Veyrat C, Chiche P. Right venous return registered by transcutaneous route at the level of the internal jugular vein in normal persons. Physiologic interpretation of the curves. *Bull Mem Soc Med Hop Paris* 1968; 119: 873–889.
31. Matsuo H, Nimura Y, Kitabatake A, et al. Analysis of flow patterns in blood vessels with the directional ultrasonic Doppler technique through a transcutaneous approach. *Jpn Circ J* 1973; 37: 735–746. DOI: 10.1253/cj.37.735
32. Benchimol A, Harris CL, Desser KB. Noninvasive diagnosis of tricuspid insufficiency utilizing the external Doppler flowmeter probe. *Am J Cardiol* 1973; 32: 868–873. DOI: 10.1016/s0002-9149(73)80020-7
33. Sivaciyan V, Ranganathan N. Transcutaneous Doppler jugular venous flow velocity recording. *Circulation* 1978; 57: 930–939. doi:10.1161/01.CIR.57.5.930
34. Appleton CP, Hatle LK, Popp RL. Superior vena cava and hepatic vein Doppler echocardiography in healthy adults. *J Am Coll Cardiol*. 1987; 10: 1032–1039. doi:10.1016/S0735-1097(87)80343-1
35. Pennestrì F, Loperfido F, Salvatori MP, et al. Assessment of tricuspid regurgitation by pulsed Doppler ultrasonography of the hepatic veins. *Am J Cardiol*. 1984; 54: 363–368. doi:10.1016/0002-9149(84)90198-X
36. Alzeer A, Arora S, Ansari Z, et al. Central venous pressure from common iliac vein reflects right atrial pressure. *Can J Anaesth* 1998; 45: 798–801. DOI: 10.1007/bf03012153
37. Abu-Yousef MM, Kakish M, Mufid M. Pulsatile venous Doppler flow in lower limbs: Highly indicative of elevated right atrium pressure. *AJR Am J Roentgenol*. 1996; 167: 977–980. doi:10.2214/ajr.167.4.8819397
38. Abu-Yousef MM. Normal and respiratory variations of the hepatic and portal venous duplex Doppler waveforms with simultaneous electrocardiographic correlation. *J Ultrasound Med*. 1992; 11: 263–268. doi:10.7863/jum.1992.11.6.263
39. Ranganathan N, Sivaciyan V, Pryszyk M, et al. Changes in jugular venous flow velocity after coronary artery bypass grafting. *Am J Cardiol* 1989; 63: 725–729. DOI: 10.1016/0002-9149(89)90259-2
40. Prager R, Pratte M, Kenny JE, et al. A wireless, wearable carotid Doppler ultrasound aids diagnosis and monitoring of pericardial tamponade: A case report. *Crit Care Explor* 2023; 5: e0911. DOI: 10.1097/cce.0000000000000911
41. Matsuyama W, Ohkubo R, Michizono K, et al. Usefulness of transcutaneous Doppler jugular venous echo to predict pulmonary hypertension in COPD patients. *Eur Respir J* 2001; 17: 1128–1131. DOI: 10.1183/09031936.01.00088401
42. Burgess MI, Bright-Thomas R. Usefulness of transcutaneous Doppler jugular venous echo to predict pulmonary hypertension in COPD patients. *Eur Respir J* 2002; 19: 382–383.
43. Kenny J-ES. Wearable ultrasound for continuous deep-tissue monitoring. *Nat Biotechnol*. 2024. DOI: 10.1038/s41587-023-02098-8
44. Kenny J-ÉS, Munding CE, Eibl AM, et al. Wearable ultrasound and provocative hemodynamics: A view of the future. *Crit Care*. 2022; 26: 329. DOI: 10.1186/s13054-022-04206-7
45. Kenny JS, Eibl JK, Horner C, et al. Hemodynamic insights from simultaneous common carotid and internal jugular Doppler ultrasonography in a patient with hypoxemia and multiple organ dysfunction. *Chest* 2024; 165: e107–e112. DOI: 10.1016/j.chest.2023.11.023
46. Kenny J-ÉS, Barjaktarevic I, Mackenzie DC, et al. Inferring the Frank–Starling curve from simultaneous venous and arterial Doppler: Measurements from a wireless, wearable ultrasound patch. *Front Med Technol* 2021; 3. Hypothesis and Theory. DOI: 10.3389/fmedt.2021.676995
47. Kenny JS, Gibbs SO, Eibl JK, et al. Simultaneous venous-arterial Doppler during preload augmentation: Illustrating the Doppler Starling curve. *Ultrasound J* 2023; 15: 32. DOI: 10.1186/s13089-023-00330-9
48. Kenny J-ÉS, Prager R, Rola P, et al. Simultaneous venous-arterial Doppler ultrasound during early fluid resuscitation to characterize a novel Doppler Starling curve: A prospective observational pilot study. *J Intensive Care Med*. 2024: 08850666231224396. DOI: 10.1177/08850666231224396

49. Kenny J-ES. Assessing fluid intolerance with Doppler ultrasonography: A physiological framework. *Med Sci (Basel)*. 2022; 10: 12. doi:10.3390/medsci10010012
50. Kenny J-ES. Passive leg raising uncovers venous congestion: Dynamic fluid intolerance and the Doppler Starling curve. *Crit Care*. 2024; 28: 388. DOI: 10.1186/s13054-024-05171-z
51. Kenny JS, Prager R, Rola P, et al. Unifying fluid responsiveness and tolerance with physiology: A dynamic interpretation of the Diamond-Forrester classification. *Crit Care Explor* 2023; 5: e1022. DOI: 10.1097/cce.0000000000001022
52. Kattan E, Castro R, Miralles-Aguiar F, et al. The emerging concept of fluid tolerance: A position paper. *J Crit Care* 2022; 71: 154070. DOI: 10.1016/j.jcrc.2022.154070
53. Kenny JS, Rola P. Optimising flow without congestion using the venous-arterial Doppler enhanced resuscitation framework. *Australas J Ultrasound Med* 2024; 27: 193–196. DOI: 10.1002/ajum.12388
54. Nunnally J, Ko SM, Ugale K, et al. A nursing-led sepsis response team guiding resuscitation with point-of-care ultrasound: A review and model for improving bundle compliance while individualizing sepsis care. *SAGE Open Med*. 2024; 12: 20503121241290378. DOI: 10.1177/20503121241290378
55. Kenny J-ES, Prager R, Rola P, et al. The effect of gravity-induced preload change on the venous excess ultrasound (VExUS) score and internal jugular vein Doppler in healthy volunteers. *Intensive Care Med Exp*. 2023; 11: 19. DOI: 10.1186/s40635-023-00504-8
56. Lipton B. Estimation of central venous pressure by ultrasound of the internal jugular vein. *Am J Emerg Med*. 2000; 18: 432–434. doi:10.1053/ajem.2000.7335
57. Mulder TA, Becude L, Lopez Matta JE, et al. The SONIC CENTRAL study: Central venous pressure in hospitalized patients – Accuracy of physical examination and point-of-care ultrasonography of the jugular vein and inferior vena cava. *CHEST Critical Care* 2024; 2: 100091. DOI: <https://doi.org/10.1016/j.chstcc.2024.100091>
58. Kenny J-ES, Horner C, Elfarnawany M, et al. A wireless, wearable ultrasound for assessing left ventricular assist device hemodynamics: A case series. *SN Compr Clin Med* 2024; 7: 5. DOI: 10.1007/s42399-024-01769-0
59. Kenny J-ES. A framework for heart-lung interaction and its application to prone position in the acute respiratory distress syndrome. *Front Physiol*. 2023; 14. Hypothesis and Theory. DOI: 10.3389/fphys.2023.1230654
60. Persichini R, Lai C, Teboul JL, et al. Venous return and mean systemic filling pressure: Physiology and clinical applications. *Crit Care* 2022; 26: 150. DOI: 10.1186/s13054-022-04024-x